

 Bio 431: Unknown Bacterial Identification — Data Sheet

USAFA Department of Biology · Operational Microbiology

CADET NAME C2C First Last	LAB SECTION T1
CULTURE NUMBER Unknown #1	DATE 24 March 2026

 **Incident Report: Operation Sandstorm Sweep**

Al Udeid Air Base, Qatar. Following a sustained sandstorm that grounded operations for 72 hours, maintenance personnel discovered a thick, mucoid growth coating the interior surfaces of the base water purification system's filtration membranes. The growth appeared dry and powdery on exposed surfaces but slimy where moisture was present. Three airmen who consumed water from the affected system reported mild gastrointestinal distress. The base commander has ordered an immediate investigation to determine if the water supply is safe for consumption or if the system has been compromised by a pathogenic organism.

Source: Water purification system filtration membrane | Location: Al Udeid Air Base, Qatar | Sample ID: SS-WP-001

 **Final Identification — Confirmed**

Bacillus subtilis

BSL-1 | Confidence: High

MORPHOLOGICAL CHARACTERISTICS

Gram Reaction	Positive
Cell Shape	Rod
Arrangement	Single / Chains
Endospore	Positive
Acid-Fast	Negative
Capsule	Not observed

CULTURAL CHARACTERISTICS

Colony Size	Medium (2–4 mm)
Colony Shape	Irregular
Margin	Undulate
Elevation	Flat to raised
Pigment	None (off-white/cream)
Opacity	Opaque
Texture	Dry, rough
Growth Medium	Trypticase Soy Agar

BIOCHEMICAL / PHYSIOLOGICAL CHARACTERISTICS

TEST	RESULT	DATE RECORDED
Catalase	Positive	3/17/2026
Oxygen Requirements (FTM)	Obligate Aerobe	3/17/2026
Endospore Stain (Schaeffer-Fulton)	Positive	3/17/2026
Voges-Proskauer (VP)	Positive	3/19/2026
Gelatinase	Positive	3/19/2026

TEST	RESULT	DATE RECORDED
Amylase (Starch Hydrolysis)	Positive	3/19/2026
Nitrate Reduction	Positive	3/21/2026
Motility	Positive	3/21/2026